Phase 3H | First-coach parts and connections

PULSIPHER FARMS / MOAPA VALLEY CORN MAZE

Dimensions that lead to build details

PF-TRAIN-S3H-R0 | 18 October 2026 UTC

Preliminary shop-drawing development. Not released for fabrication, vehicle testing or passenger operation.

This issue responds to the request for more measurements, sizes, materials and an explanation of how parts connect. It develops the first regular coach in sequence: retained steel frame, floor panels, seat-support pattern and worked joint examples. Earlier issues remain preserved. New dimensions describe specific study parts and interfaces; they do not replace qualified design acceptance.

Now defined for review	Evidence needed to release
110 frame pieces with exact IDs, nominal sections, orientation, geometric lengths and end coordinates; grouped material and stock schedules.	Certified material/stock, accepted vehicle loads, end preparations, joint gaps, final tolerances and responsible engineer acceptance.
Located frame contacts and enlarged calculated crossmember/seat-rail connection examples.	Complete local-wall, fatigue, load-reversal, weld-access and inspection design; primary bolster, bogie and coupler attachments.
Floor-panel cutting/support study and H-S16 seat-support coordinates tied to retained rails.	Qualified panel span/point-load behavior, closed edges, hardpoints, attachments and occupied interior approval.
Selectable part model, exploded views, dimensioned sheets, editable cut/connection registers and calculations.	Released manufacturing drawings, accepted procedures and actual inspection records.

An envelope length is the nominal distance between modeled member faces. It is not automatically a saw length after root gaps, coping, weld preparation or actual tube tolerances. Calculated joint trials are separate from released capacities. Unresolved holes, pins, plates and procedures are not filled with guessed sizes.

01 | Retained basis, changes and datums

H1 retains all 110 members of the F1 frame and its E1 32-inch-floor integration study. It does not select a new floor height over the earlier 28-inch baseline. Interior dimensions, seating positions, 36-inch gauge evaluation, 0% longitudinal-grade target and deferred North Pole location remain unchanged. CR-G01 remains current: air service brakes with spring-applied, air-released parking; commonly sourced combination chambers are candidates, not selected assemblies.

Reference	Dimension / convention
Coordinates	Inches unless stated. x starts at the front inside end wall and increases rearward. y is transverse about centreline. z is above top of rail. Forward travel is toward decreasing x.
Frame envelope	x = -47 to 353; y = -55 to 55. Overall 400 × 110 inches. Cabin outer length 312; interior clear length 306 and width 104 inches.
Bogie centres	x = 45 and 261; spacing 216 inches. Retained packaging locations, not completed attachments.
Datum A	Top of retained steel: z = 31.003937 inches. Nominal floor stack reaches the 32-inch study finished floor.
Datum B / C	B is y = 0 from main-girder centre lines y = ±34. C is forward frame face x = -47; cabin x = 0 lies 47 inches aft.
Check dimensions	The 216 × 68-inch rectangle at the paired girder/bogie checkpoints has theoretical diagonal 226.450878 inches. The outer 400 × 110 envelope diagonal is 414.849370 inches. These are not acceptance tolerances.

Retained: F1 steel geometry and material-property inputs. **Revised study:** H-S16 foot-centre offsets avoid earlier points on member faces while seats stay in place. **Rejected assumptions:** treating every contact as the same joint; accepting shear-only welds as complete joints; transferring seat loads through unqualified honeycomb skins.

This is a focused first-regular-coach package. Accessible coach, locomotive, complete body shell, couplings, stairs, brake installation, track and station construction drawings are not claimed complete. Their governing interfaces and earlier review status remain open.

02 | Frame parts, materials and connections

PF-TRAIN-S3H-R0 · 18 October 2026 · **Preliminary engineering development, not production instructions**

What is now measured and indexed

The package retains every one of the 110 F1 frame members and the 32 in finished-floor study. frame_h.json gives each member's ID, section, orientation, centre, bounding faces, end-centre coordinates, envelope length and adjoining connection IDs. The geometry remains 400 × 110 in overall, from x−47 to 353 and y−55 to 55. The steel bearing plane is z31.003937 in above rail. Main girder centres are y±34; perimeter rail centres are y±54. Bogie/bolster stations remain x45 and x261, 216 in apart. Grade remains 0%; CR-G01 retains air-service braking with spring-applied, air-released parking. No chamber mounting holes are introduced.

RETAINED means unchanged study geometry. **VERIFIED** identifies a published catalog property. **STUDY** identifies a proposed material, arrangement or calculation assumption. **HELD** identifies a missing production decision. A zero nominal end-plane gap is a geometric contact, not an approved welding root opening.

Exact frame schedule

All dimensions below are inches. Section depth is vertical; the first section dimension is not rotated horizontally. The table contains all 110 members, including the segmented seat rails.

Section, nominal outside dimensions	Envelope length	Quantity	Function
HSS 12×4×1/4	400	2	Continuous main girders
HSS 12×4×1/4	64	2	Centre bolster segments
HSS 12×4×1/4	17	4	Outer bolster segments
HSS 4×2×1/8	400	2	Perimeter rails
HSS 4×2×1/8	64	16	Ordinary centre cross members
HSS 4×2×1/8	17	32	Ordinary outriggers
HSS 3×1×1/8	22	36	Seat attachment rails
HSS 3×1×1/8	21	16	Seat rails adjoining deep bolsters

The schedule is 4492 in, or 374 ft 4 in, of net envelope length. Catalog-based bare steel mass remains 1527.288 kg. Connection steel, weld metal, caps, coatings, bogies and body equipment are additional. The 400 in members require 33 ft 4 in uninterrupted study lengths; no longitudinal splice is detailed. Each cut-group row lists the exact member IDs, preventing the 21 and 22 in seat segments from being interchanged.

Atlas's current manufacturer-hosted table confirms nominal weights 25.82, 4.75 and 3.05 lb/ft respectively. Its design walls are 0.233, 0.116 and 0.116 in; these differ from nominal purchase thicknesses. The corresponding strong-axis I values are 119, 2.65 and 0.817 in^4. These are catalog properties, not measurements of delivered steel. [C3HS01]

The material callout is **ASTM A500/A500M-23 Grade C rectangular HSS, study candidate**, using Fy 50 ksi and Fu 62 ksi. Heat identification, MTR, delivered wall/corner measurements and project toughness acceptance remain required. ASTM's public scope specifically cautions that A500 may not suit dynamically loaded welded elements where notch toughness matters. The engineer should resolve that issue before purchasing; an A1085 alternative requires verified size availability and recalculated properties rather than an unrecorded substitution. [C3HS02-03]

Purchasing and cut planning

Two auditable nesting scenarios assign all 110 IDs to stock bars. They assume 1/8 in kerf per piece and 1/2 in trim at each stock end, solely for purchasing comparison. Actual saw cuts, root gaps, shrinkage allowances, end preparation and tolerances remain blank.

The 40 ft-only first-fit-decreasing plan uses 12 bars: three G12, six C4L and three SR. It purchases 480 ft and 2088.521 kg by catalog weight. Reducing lightly used bars to 20 ft gives nine 40 ft plus three 20 ft bars, 420 ft and 1783.525 kg. That mixed plan has 13.75 in kerf, 12 in end trims and 522.25 in remaining offcuts. It saves 60 ft and 304.996 kg of purchased stock against the all 40 ft scenario. These are reproducible heuristic patterns, not a claim of optimal nesting or supplier stock. Final joint gaps or an alternate material can change them.

Located connection families

The contact map accounts for 212 connected member ends and eight free ends. It distinguishes 32 ordinary centre-to-main joints, 32 outrigger-to-main joints, 32 outrigger-to-perimeter joints, eight bolster-to-main joints, four bolster-to-perimeter joints, 88 seat-to-ordinary-crossmember joints and 16 seat-to-bolster joints.

The four deep bolster ends meet the 4 in-deep perimeter rail over only 4 of their 12 in depth. The remaining 8 in has no matching perimeter face. These require a dedicated transition/load-path detail; extending an ordinary crossmember weld around the entire deep end is physically unsupported. The map reports outer-envelope contact rectangles, not actual metal bearing area. Bolster/main, bogie support, coupler/draft, lifting/jacking and equipment interfaces remain separate engineering packages.

Worked crossmember joint CM-01

This detail locates **T93_1 to G1**, at y32, x92-94 and z27.003937-31.003937. The member spans 64 in between main-girder faces. Its example loading is 4.8 kPa floor live pressure plus an assumed 1.0 kPa dead pressure over 24 in tributary width, plus member selfweight. The crowd-pressure case does not also add seated passenger loads. These assumptions do not replace the vehicle's global load cases.

The gravity reaction is 2.930 kN. An explicit 2× gravity sensitivity gives 5.860 kN shear. A fully restrained end gives 1.588 kN·m; the calculation also shows zero and 50% restraint. Face loading adds 0.298 kN·m about the receiver girder axis through its 2 in offset, requiring global torsion/local-shell assessment. Simple-span gravity bending is 54.64 MPa with 1.486 mm deflection, before the sensitivity multiplier.

The located **trial** has two vertical 1/4 in fillets, each 3 in long, from z27.378937 to30.378937. Their XY cross-sections are supplied as exact triangular prisms for drawings. They leave the top bearing surface unwelded. Their vertical limits fall inside both catalog workable-flat bands; actual corners and access still need measurement.

The elastic weld-only calculation uses throat=leg/√2, Aw=2aLe, Iw=2aLe³/12, τ=V/Aw and σ=M(Le/2)/Iw. The vector stress is compared with 0.75×0.60×70 ksi, without directional enhancement. These reference coefficients are documented by STI; they are not an adopted vehicle load or welding code. [C3HS04-05]

At full restraint, the trial gives 182.90 MPa against 217.18 MPa, ratio 0.842. Using 3/16 in legs gives 1.123;1/8 in gives 1.684. Thus a shear-only check would conceal a meaningful restraint issue. The branch-wall load-strip sensitivity reaches 279.34 MPa against a candidate 310.26 MPa yield comparison.

This is not local HSS connection capacity. Receiver-wall bending, plastification, punching, eccentric root stresses, effective length and shell interaction remain unresolved.

Worked seat-rail joint SR-01

This detail locates **S20_3 to T69_1**, at x70, y19.5-20.5 and z28.003937-31.003937. The rail has 22 in face span. The coordinated H-S16 mounting study retains the seat row at x73 but places its rear foot at x81, centred in this span. The four feet use x±8 and y20/44 relative support lines; this changes the prior foot arrangement without changing any frame member.

Two 145 kg adults plus a 15 kg bench give 305 kg and 747.757 N per foot under the stated symmetric distribution. Rail selfweight adds 24.873 N. The end reaction is 386.315 N; the 2× sensitivity is 772.630 N with 106.778 N·m fully fixed end moment. The receiver-axis offset contributes 19.625 N·m torsion.

Two 3/16 in trial side fillets, each 1.5 in long, occupy z28.753937-30.253937. Their weld-only full-restraint ratio is 0.302; the base-wall strip sensitivity is 75.13 MPa. This does not approve seat anchorage, passenger inertia, abuse, pull-out, torsional or fatigue loading.

Turning these details into fabrication documents

Both trial weld legs exceed the 1/8 in nominal branch walls. More deposited weld metal cannot strengthen an insufficient tube wall or establish fatigue resistance. A welding engineer must establish permissible detail size, effective length, process, heat input, burn-through control and qualification. The responsible vehicle engineer must establish joint restraint, shell reinforcement needs and cyclic load/stress ranges. AWS currently lists D15.1/D15.1M:2025 for cars, locomotives and components; its application and inspection basis must be deliberately selected. [C3HS06]

The next practical package is a released joint schedule with actual end gaps/preparation, material acceptance, WPS references and inspection points. Farm staff can reconcile IDs, inventory material and prepare the marked layout/fixture under the approved work package. Structural cutting, tacking and welding follow that release. The current files provide exact review geometry and calculated trials; production saw lengths, weld selection and acceptance tolerances remain explicitly unissued.

03 | Floor panels and seat-to-frame path

Dimensioned fabrication development, 18 October 2026. Not for fabrication. This module advances the first regular coach with two explicit trials: **H-P125**, a stock-sheet floor layout, and **H-S16**, a revised seat-foot arrangement. Neither alters the retained F1 frame: 110 members, 306 × 104-inch clear cabin, 312 × 110-inch outer floor and the named 32-inch floor-height study. Previous drawings remain preserved. The current air-brake direction, CR-G01, creates no additional floor or equipment approval here.

Panel parts and supported joints

H-P125 divides the floor at x = −3, 45, 93, 141, 189, 237, 285 and 309 inches; longitudinal seams are y = ±34 inches. The trial uses **1/8-inch internal gaps and 1/16-inch perimeter inset**. These are stated fit-up assumptions for supplier review, not established thermal-joint requirements. All dimensions below are **finished panel envelopes**, including whatever edge closure is subsequently engineered; they are not raw honeycomb-core cutting instructions.

Panel IDs	Quantity	Finished envelope, inches
HP01L/R through HP06L/R	12	47.875 × 20.875 × 0.750
HP01C through HP06C	6	47.875 × 67.875 × 0.750
HP07L/R	2	23.875 × 20.875 × 0.750
HP07C	1	23.875 × 67.875 × 0.750

The script checks the actual steel top faces at every transverse boundary: their rectangular-envelope union covers y = −55 to +55 without inserting additional supports. Ordinary seams sit over 2-inch-wide crossmembers; x45 has a 4-inch-wide bolster. Each ordinary seam has **0.9375 inch / 23.8125 mm** bearing per panel after the trial gap. The 4-inch girders at y±34 provide **1.9375 inches / 49.2125 mm** per side. The outer panel edge at y54.9375 retains the same nominal bearing on perimeter rail y53…55. These are gross rectangular overlaps. Rounded HSS corners, welds, closeouts and actual flatness reduce usable contact; effective top-flat widths remain unknown.

A nine-sheet 48 × 96-inch nest includes a **1/8-inch trial saw kerf**, all 21 pieces and no overlaps. It does not reserve factory-edge trimming or establish minimum usable sheet size. Center pieces orient their 67.875-inch dimension along the 96-inch stock length. The 47.875-inch cross dimension leaves only 1/8 inch total spare across the 48-inch sheet; two-up side pieces likewise leave 1/8 inch along 96 inches; an OEM finished-panel quotation is preferable to assuming nominal sheets can be trimmed arbitrarily. Bare panel mass is **100.99 kg**, excluding closeouts, inserts, joint material and waste. The corresponding zero-gap outer envelope is 101.62 kg; this is not an additional coach-weight saving against the earlier mass allocation.

The candidate Plascore AA3.6-80 has 0.020-inch primed aluminum skins, aluminum honeycomb and toughened-epoxy bonding. The 0.75-inch version is published at 0.94 lb/ft². Standard sheets have open edges; factory closeouts and inserts are available. Supplier-return dimensions must include closed-edge construction, skin orientation, tolerances, caster qualification and bearing requirements before cutting. [H-B01, H-B02]

Floor-to-steel section

The bottom-to-top section is fixed for coordination:

Layer	Nominal thickness	Top elevation above rail
Existing F1 steel top	-	31.003937 in
Bedding/isolation reservation	3.00 mm	31.122047 in
Sandwich panel	19.05 mm	31.872047 in
Finish-adhesive reservation	1.00 mm	31.911417 in
Sirius NT finish	2.25 mm	32.000000 in

The 3 mm layer is not a selected rubber strip. Sikaflex-252 is an electrically nonconductive elastic vehicle adhesive worth qualifying, but its typical 2.5 MPa lap-shear value is **not** a floor-bond allowable or compression/creep rating. The 1 mm finish layer is only a packaging reservation. [H-B05]

At the retained 4.8 kPa comparison load and 24-inch support spacing, one panel-edge reaction is **1.463 kN/m**. Dividing by the nominal bearing gives **61.44 kPa** at an ordinary seam, or **29.73 kPa** at a bolster. These full-gross-contact averages are lower-bound contact demands for the stated reaction: actual local pressure may be higher. The screen excludes caster loads and joint flexibility. Using Atlas catalog workable-flat widths instead gives **16.6751/34.925 mm** per-panel contact and **87.74/41.89 kPa** for ordinary/bolster seams. These are separate catalog sensitivities, still requiring measured contact. [H-B09] The retained mobility-device sensitivity gives **3.273 kN at one wheel**; its footprint and actual floor response remain unknown.

A soft gasket cannot simply replace the bedding reservation: Rogers publishes 35 kPa at 25% compression for PORON 4701-30-20, below the ordinary-seam pressure screen. That comparison is not a complete foam model, but demonstrates why a compressible gasket is not an automatically interchangeable rigid packer. [H-B08]

Proposed load path H-F1/H-F2 is panel → qualified closed edge/bearing → steel, with engineered retention for uplift and in-plane forces. No screws into 0.020-inch skin are assigned structural credit. Sika discusses typically 3 mm or thicker elastic bonds, compatible spacers and humidity-dependent curing; even a 20 mm wide joint can take 1-2 weeks to cure fully under standard conditions. Consequently, the drawing does not specify a blanket 2- or 4-inch-wide adhesive bed or a production cure interval. [H-B06]

Movement studies impose 0.5, 1 and 2 mm relative displacement across a nominal 3 mm bond, giving engineering shear strains 0.167, 0.333 and 0.667. These are demands, not permitted movements or temperature predictions. Sealant depth, fatigue, water exposure, coatings and cure access need joint qualification. Sikaflex-221 is an identifiable sealing candidate, subject to substrate testing. [H-B07]

Bench-to-steel connection

The earlier ±6-inch foot references put **eight of 64 feet exactly on member faces** at x47 or 259. H-S16 changes only the longitudinal mounting reference to **row center ±8 inches**, making a symmetric **16 × 24-inch foot pattern**. All row centers, 40 × 20-inch bench envelopes and y±20/±44 seat rails stay in place. All 64 revised centers lie within actual rail segments; the minimum distance to a member face is 2 inches. This establishes point placement, not usable bolt edge distance or a baseplate envelope.

For bench HB03R, row 105, the four references are **(97,20), (97,44), (113,20), (113,44)** inches. They connect to S20_4/S44_4, each HSS 3 × 1 × 1/8, extending x94…116. The proposed physical sequence is bench pedestal/base → engineered rigid hardpoint → locally reinforced HSS rail → each rail’s end joints → crossmember → main girders. The thin floor skin and interior wall panels carry no seat-anchor load.

H-S1 gives an exploded section. The known floor-to-steel separation is **25.30 mm**. A rigid load bypass crosses that stack through a sealed clearance opening; any reinforcing cap above steel consumes part of that height. Its diameter, reinforcement, bolt group, welds, torque and access remain unselected. A single bolt squeezing honeycomb is not this connection. The 1-inch rail has only a **0.438-inch catalog workable**

top flat; closed HSS access and a properly reinforced interface require coordinated design; a point marker cannot justify a drill hole.

With two 145 kg adults and a 15 kg bench allowance, total mass is 305 kg. Gravity produces **0.748 kN per foot**; a 1.5 vertical multiplier gives 1.122 kN. Individual rail reactions are calculated between their physical end faces, without assuming continuity through a crossmember.

A separate longitudinal 1g sensitivity, combined CG 24 inches above the floor, gives 2.991 kN horizontal force and 1.823 kN·m overturning. Under equal four-foot sharing, each foot has 0.748 kN direct shear; gravity restores 0.748 kN while overturning adds/subtracts 2.243 kN. The high reaction is **2.991 kN compression**, the low reaction **−1.496 kN**, requiring uplift resistance. A separate lateral 0.5g case gives 1.496 kN high compression and zero low reaction. These are analyst sensitivities, not adopted seat-restraint cases; occupant/back forces, simultaneous axes and fatigue must be established by the responsible engineer.

Finish and first physical work

Two Sirius NT rough blanks, **7872.4 ×1420.8 mm**, cover the clear floor in halves with 50 mm trimming allowance at each end and edge. The candidate finish seam is y0, distinct from structural seams. Gerflor specifies 2.25±0.15 mm NT thickness, 200 cm rolls, a 1±0.5 mm finish factory-edge gap and 5 mm CR50 flat-weld cord. Its subfloor must be secure, flat and smoothly filled; adhesive compatibility remains essential. A continuous finish symbol over a panel seam is not proof that it can accommodate that seam’s movement. [H-B03, H-B04]

The farm crew can lay out these coordinates and check ground-level templates now. Loaded seam and seat prototypes require approved fixtures and test criteria. The located seam demonstrator spans x45…141/y±34; the seat demonstrator uses HB03R and its real rail segments. Record panel-relative and support movement, joint opening, water ingress, hardpoint uplift, loosening and permanent deformation. No physical results are claimed. Production release follows accepted panel/edge details, measured stack, engineered hardpoints and approved inspection procedures.

Source URLs, dated verification and limitations are in **body_h.json**; calculations and the full cut-envelope/coordinate schedule are in **body_h_calculations.json**, reproducible with **body_h_calc.py**.

04 | Assembly sequence and inspection

This traveler defines the intended sequence after individual design releases exist. It preserves part identity, prevents concealment of uninspected work and records changes. It does not authorize cutting or welding from this study.

H-F01 | Reconcile source revision and part identity

Record: 110 frame IDs, current section/material candidates, drawing revision and connection family.

Farm crew: Print labels; maintain parts/heat register. Specialist: Release final part/joint drawings.

Start condition: Geometry and connection reviews complete.

H-F02 | Receive and inspect structural tubes

Record: Mill certificate/heat, outside dimensions, wall measurement method, bow, twist, seam and damage.

Farm crew: Receive, photograph and quarantine unmatched stock. Specialist: Approve material/toughness and accept stock.

Start condition: Selected specification and current mill evidence.

H-F03 | Survey fixture and frame datums

Record: Top steel plane A; centreline B; front face C; paired bogie and perimeter checkpoints.

Farm crew: Assist measurements using issued procedure. Specialist: Design support/clamp layout and set acceptance tolerances.

Start condition: Fixture design and calibrated instruments.

H-F04 | Mark and prepare individual members

Record: Part ID, heat link, section orientation, geometric length, approved end preparation and actual length.

Farm crew: Mark/handle parts after instruction. Specialist: Release cutting, joint gaps and any machine preparation.

Start condition: Cut envelope converted into an approved finished part.

H-F05 | Dry assemble primary frame

Record: Two main girders, bolster segments, perimeter; no assumed continuous transverse beam through the girders.

Farm crew: Assist setup under fabrication supervisor. Specialist: Fit joints, verify access and inspect geometry.

Start condition: Designed joints and fixture.

H-F06 | Fit crossmembers and seat-support rails

Record: Every member ID and joint occurrence; straightness, top plane, floor-seam bearing and seat hardpoint access.

Farm crew: Check labels and measurement records. Specialist: Resolve any forced fit or gap before welding.

Start condition: Connection map and dimensional criteria.

H-F07 | Weld and inspect in controlled stages

Record: Joint IDs, accepted WPS, welder ID, applicable procedure parameters, distortion and required inspection results.

Farm crew: Maintain traveler; do not conceal before inspection. Specialist: Qualified welding/inspection, repair disposition.

Start condition: Project welding basis, joint design and qualifications.

H-F08 | Trial-fit floor edges and seat hardpoints

Record: Panel dimensions/gaps, seam support, edge closure, solid load path to steel, underside tool access.

Farm crew: Dry-fit templates under instruction. Specialist: Approve inserts/attachments and panel qualification.

Start condition: Exact panel/adhesive/seat anchorage design.

H-F09 | Protect and close assemblies

Record: Surface preparation/coating, isolation and sealed edges; compatibility and inspection access.

Farm crew: Approved noncritical preparation/finish work. Specialist: Approve coatings/bonding details and hidden-work record.

Start condition: Inspection accepted before concealment.

H-F10 | Record first article

Record: As-built dimensions, component/assembly masses, deviations, controlled revision and test-release status.

Farm crew: Maintain records and photographs. Specialist: Accept geometry, load-path evidence and authorized next testing.

Start condition: No occupied test or production duplication by this issue.

The farm crew can maintain labels, photographs and material records, handle parts under an issued procedure, and assist with templates and measurements. Structural welding, wheelsets/bogies, coupler attachments, brake attachments and their inspection require the designated qualified fabricators and engineers. Any expanded crew fabrication scope must appear in released instructions, qualifications and inspection plans.

Purchased-tube tolerances and assembled-vehicle tolerances serve different purposes. The vehicle engineer must set frame straightness, twist, plane, diagonal and interface limits. This issue identifies measurement datums; acceptance values and actual measurements remain blank. Fit-up must not force distorted members into position without assessing the resulting distortion and loads.

The welding specialist must select the applicable rolling-stock welding basis, issue joint-specific procedures and qualifications, and define inspection and repair acceptance. A nominal weld leg in a study does not constitute an approved WPS.

05 | Drawings, model and editable schedules

Deliverable	Use
Dimensioned drawing set	Read member locations/IDs with cut and material schedules, enlarged joint sections and floor/seat sheets. Do not scale unknown hardware from an image.
Part inspection model	Select a member for section, length, material candidate, coordinates and joint records. Exploded offsets separate groups for visibility; they are not assembly travel or installation clearance.
Connection studies	Worked local joints relate demand, trial weld geometry and receiving walls. Passing a weld-metal screen does not qualify the complete joint or coach.
Editable workbook	Part registers, grouped stock formulas and contact records. Stock, kerf and trim are planning inputs, not verified offers. Actual inspection and approval remain blank.
Technical source package	JSON inputs, portable calculation/model scripts, SVG/DXF drawings, STEP geometry, PDF/XLSX outputs, dated sources and scoped verification records.

STEP solids use millimetres; web model data uses inches. DXF files declare units and distinguish paper-sheet layouts from actual-size details. Portable geometry does not include a native CAD assembly with complete manufacturing features and approved processes. The model carries no implied rating.

Digital file/geometry checks and document rendering are separate from engineering acceptance. Browser execution could not be observed because the supported managed preview browser was unavailable. Static views and a selectable plan fallback are provided. No physical fabrication, load test or approval is claimed.

06 | Next controlled development

Continue with the first-coach structural connections, using this part/joint register as the index. Resolve ordinary crossmember and seat-rail trials, including effective weld length, local tube walls, restraint and fatigue; then complete primary bolster-to-main/perimeter and body-to-bogie load paths. Coordinate floor seams and seat hardpoints before closing the floor.

Required completion	Concrete return
Loads and materials	Responsible engineer’s vehicle load combinations/fatigue basis, selected material/toughness criteria and certified available stock.
Connections	Resolved wall limit states, weld/fastener details, exact plates/holes/gaps/preparation, tool access, tolerances and inspection criteria.
Bogie and brakes	Actual manufacturer geometry, ratings and motion; body support/retention; coordinated air chambers, foundation brake, reservoirs, controls and failure response.
Floor and seats	Qualified panel span/point-load data, edge/seam system, hardpoint/baseplate geometry and transfer into verified steel members.
First article	Released scope/drawings, material and WPS records, inspection traveler and empty/ballasted test plan before passenger use.

Safety-critical unresolved dimensions are identified by item. This package advances part definition and calculations while leaving engineering responsibility, design acceptance and physical qualification explicit.

07 | Dated primary-source register

Primary references checked on 18 October 2026 UTC. Catalog data is not a stock confirmation, quotation, certified heat, application approval or fabrication release. No supplier contact or order was made.

C3HS01 | Atlas Tube A500 rectangular HSS section properties

Accessed: 2026-10-18

Rows HSS 12x4x1/4, 4x2x1/8 and 3x1x1/8 rechecked: nominal walls, design walls, weight, I/S and workable flats.

Limit: Catalog table is not stock, material certification or actual corner geometry.

https://www.atlastube.com/wp-content/uploads/2018/04/Atlas-HSS-Section-Properties-A500-Rectangles.pdf

C3HS02 | ASTM A500/A500M-23 public scope

Accessed: 2026-10-18

Public scope identifies structural tubing and cautions regarding dynamically loaded applications where notch toughness matters.

Limit: Full standard and vehicle material suitability require responsible engineer review.

https://store.astm.org/a0500_a0500m-23.html

C3HS03 | Steel Tube Institute material specification comparison

Accessed: 2026-10-18

A500 Grade C material candidate and distinction from A1085 including notch toughness; 50 ksi Fy and 62 ksi Fu are candidate inputs.

Limit: No actual heat certification, exact stock or rail application approval.

https://steeltubeinstitute.org/resources/understanding-hss-material-specifications-which-astm-should-i-specify-for-hss/

C3HS04 | Steel Tube Institute Full-Capacity Welds

Accessed: 2026-10-18

2024 article gives phi_weld = 0.75 and Fnw = 0.60 FEXX and compares weld/base-wall strength.

Limit: General HSS design reference; not adopted vehicle welding specification or fatigue acceptance.

https://steeltubeinstitute.org/resources/full-capacity-welds-cjp-vs-fillet-vs-fillet-reinforced-pjp/

C3HS05 | Steel Tube Institute Design of Fillet Welds to Rectangular HSS

Accessed: 2026-10-18

Effective weld lengths, local eccentricity and restrictions on directional enhancement discussed.

Limit: Older technical article; no directional increase used here and actual effective length remains to be established.

https://steeltubeinstitute.org/resources/design-of-fillet-welds-to-rectangular-hss/

C3HS06 | AWS D15 Committee on Railroad Welding

Accessed: 2026-10-18

Current committee lists D15.1/D15.1M:2025 for railroad rolling stock/components.

Limit: Listing alone does not select project welding basis, qualify WPS or assign inspection criteria.

https://www.aws.org/About/Get-Involved/Committees/D15-Committee-on-Railroad-Welding/

C3HS07 | AISC360 current standards listing and errata

Accessed: 2026-10-18

Current official listing identifies AISC 360-22; errata page lists September 2026 revisions.

Limit: Building specification is a calculation reference here, not asserted governing vehicle code; full text/errata not used as release authority.

https://www.aisc.org/aisc/publications/current-standards/aisc-360/

H-B01 | Plascore Base Panels

Accessed: 2026-10-18

AA3.6-80: 0.020-inch primed aluminum skins, aluminum honeycomb, toughened epoxy. 0.75-inch option: 0.94 lb/ft². Standard sheets 48 × 96 inches, open edges; custom shapes and larger options offered.

Limit: Typical properties only. No project floor approval, skin alloy, minimum bearing, caster capacity, production tolerance or quotation.

https://plascore.com/composite-panels/base-panels/

H-B02 | Plascore Value Add

Accessed: 2026-10-18

Factory edge closeouts, frames, inserts, machining, sealing and through-fastening options offered.

Limit: No project insert or edge-closeout rating, size or configuration.

https://plascore.com/composite-panels/value-add/

H-B03 | Gerflor Tarabus Sirius technical datasheet

Accessed: 2026-10-18

NT backing: 2.25 ± 0.15 mm thickness; 2.20 ± 0.15 kg/m²; roll width 200 cm (−0/+1 cm), length 24 m.

Limit: Candidate transport finish; published fire/slip tests do not approve the complete coach.

https://cdn.gerflor.com/media/2/38529/tarabus%20sirius%20-%20technical%20datasheet.pdf

H-B04 | Gerflor Tarabus installation guidelines

Accessed: 2026-10-18

Install at 15-30°C after 24-hour relaxation. Secure, flat subfloor with smooth filled joints. Allow at least 50 mm trim each end. Finish factory-edge gap 1 ± 0.5 mm. CR50 flat-weld cord diameter 5 mm.

Limit: Floorcovering instructions do not specify structural panel gaps or bonding to moving honeycomb joints. Adhesive compatibility and vehicle substrate acceptance remain required.

https://cdn.gerflor.com/media/2/37754/installation%20guide%20tarabus%20-%20eng.pdf

H-B05 | Sikaflex-252 United States product data sheet

Accessed: 2026-10-18

Elastic vehicle-assembly adhesive; electrically nonconductive; density 1.20 kg/L; Shore A 50; typical lap shear 2.5 MPa. Application range 10-35°C.

Limit: Not a project adhesive allowable, creep/bearing qualification or floor-finish adhesive selection. Actual substrates and conditions require tests.

https://industry.sika.com/dam/dms/us01/o/sikaflex-252.pdf

H-B06 | Sika General Guideline: Bonding and Sealing with 1-component Sikaflex

Accessed: 2026-10-18

Elastic bond layers are typically at least 3 mm, application dependent. A 20 mm wide moisture-cured joint can require 1-2 weeks for full cure at standard climate. Compatible elastic spacers and original-substrate adhesion tests are discussed.

Limit: General process guidance, not approval of 3 mm floor bedding, selected bond width, cure time or joint movement.

https://usa.sika.com/dam/dms/us01/0/General-Guideline-Bonding-Sealing-1-Component-Sikaflex-US-12-25.pdf

H-B07 | Sikaflex-221 body structure sealing

Accessed: 2026-10-18

One-component polyurethane for permanent elastic seals and simple bonding; manufacturer requires original-substrate compatibility tests.

Limit: Candidate perimeter/penetration seal only; seal dimensions, finish compatibility and movement remain unselected.

https://industry.sika.com/en/home/transportation/bus-and-coach/body-structure-sealing/sikaflex-221.html

H-B09 | Atlas HSS section properties: ASTM A500 rectangles

Accessed: 2026-10-18

Catalog workable-flat widths: HSS4 × 2 × 1/8 =1.438 inches (page5), HSS12 × 4 × 1/4 =2.875 inches (page2), HSS3 × 1 × 1/8 =0.438 inch (page6).

Limit: Catalog layout dimensions are not actual measured minimum contact widths; corners, coatings and tolerances remain to verify.

https://www.atlastube.com/wp-content/uploads/2018/04/Atlas-HSS-Section-Properties-A500-Rectangles.pdf

H-B08 | Rogers PORON 4701-30 polyurethane

Accessed: 2026-10-18

4701-30-20 thickness range 1.57-3.18 mm; typical compression force at 25% deflection is 35 kPa.

Limit: Soft gasket example screened out as an automatically interchangeable rigid floor packer; no complete floor evaluation performed.

https://www.rogerscorp.com/Elastomeric-Material-Solutions/PORON-Industrial-Polyurethanes/PORON-4701-30

H-RET01 | Frozen F1 frame and retained coach layout

Accessed: 2026-10-18

F1 has 110 modeled members, 32-inch finish datum, HSS seat rails at y ±20/±44, and retained eight seat rows.

Limit: Prior preliminary geometry and load assumptions; no fabrication release or physical survey.

H-Q01 | Steel Tube Institute HSS tolerances

Accessed: 2026-10-18

Provides measuring methods for tube dimensions and production tolerances.

Limit: Purchased tube tolerances are not the completed vehicle acceptance tolerances.

https://steeltubeinstitute.org/resources/hss-tolerances/

H-Q02 | AWS D15 Committee on Railroad Welding

Accessed: 2026-10-18

Lists current railroad welding standards and committee scope.

Limit: Project applicability and full requirements require the responsible welding/design authority; no WPS approved by this report.

https://www.aws.org/About/Get-Involved/Committees/D15-Committee-on-Railroad-Welding/